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	Document Title:	RESEARCH PROPOSAL	Revision No.	00
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(1) Research Program/Project/Study Title: Optimization of Young Coconut Multiprocessing Machine	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☐ Food Research/ ☒ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
(4) Site of implementation		
Municipality/City:		Mambusao
Province:		Capiz
Region:		VI
(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☒ Applied ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity Coconut		(8) Discipline/Field of Specialization Agricultural Machinery

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(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

Coconut is recognized as one of the most important crops in the Philippines. Producing about 347 million metric tons of fruit annually, the country is among the top producers of Coconuts in the world. The coconut (with husk) production from April to June 2023 reached 3.41 million metric tons, indicating an annual growth of 1.5 percent from the 3.36 million metric tons recorded level in the same quarter of 2022.

Coconut water known as buko juice is the liquid endosperm obtained from a young coconut which is pure and nutritious natural beverage. The sterile water, which is approximately 200 – 750 ml, is enclosed with a hard shell and a well-lined layer of coconut meat. The natural drink is high in ionic nutrients (electrolytes) such as potassium, sodium, calcium, magnesium, chloride, phosphorus and sulfur. It also contains sugar, in the form of glucose and fructose, Vitamin C, Vitamin B group and amino acids such as glutamic acid, arginine, aspartic acid, and leucine.

Young coconut is suitable for export to distant countries because the hard shell and husk provide good protection for the flesh and juice inside. The young fruit is typically trimmed to remove most of the outer husk, and is then packed in corrugated boxes for export in quantities of 9 to 10 fruits (Jarimopas & Pechsamai, 2002).

The trimming process is done manually, a task which requires skilled labor and is extremely hazardous. Currently, workers must shear the husk off the green fruit with a long sharp knife. The inner white husk is then finely shaped to form a conical shaped top, a slightly tapered cylindrical body, and a flat base. The final product has a pentagonal contour (Nobleza, et al., 2019). After it has been trimmed, the fruit is dipped into a sodium metabisulfide solution to prevent surface browning (Siwawech, 2003).

The shortage of experienced labor and the high production cost has created an urgent need for mechanical trimming machines. Many fruit traders have expressed a desire for machinery that is capable of trimming the young coconut fruit into a pentagonal shape as fast as can currently be done by skilled labor. Furthermore, they have indicated that they want the processed fruit to be free of green and fibrous areas. Finally, they require the machine to have a relatively simple design, have a low cost and to be safe (Jarimopas & Srihawong, 1996).

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned. .)

The study aims to design and develop a prototype young coconut multiprocessing machine. The main purpose of the design is to trim most of the outer husk (green husk) to create an attractive looking trimmed coconut in a hexagonal shape which could easily be cut open. The machine will also have a provision of a mechanism that pierce or drill a hole as inlet for straw and a cutting mechanism to cut the coconut into halves. Further, the equipment is designed to be used conveniently because of its simple mechanism. The equipment shall undergo test and running-in periods for adjustments and will be evaluated based on several parameters. This technology will help the coconut farmers and small-scale entrepreneurs increase their income from young coconut farming by adding value to their produce.

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

The study aims to develop an optimized coconut multiprocessing machine. Specifically, it aims to:

1. Design and fabricate a coconut multiprocessing machine that can peel young coconuts into a hexagonal shape, pierce coconut for straw inlet and cut coconut into halves;
2. evaluate the performance of the machine in terms of capacity and efficiency; and
3. perform economic analysis in operating the machine.

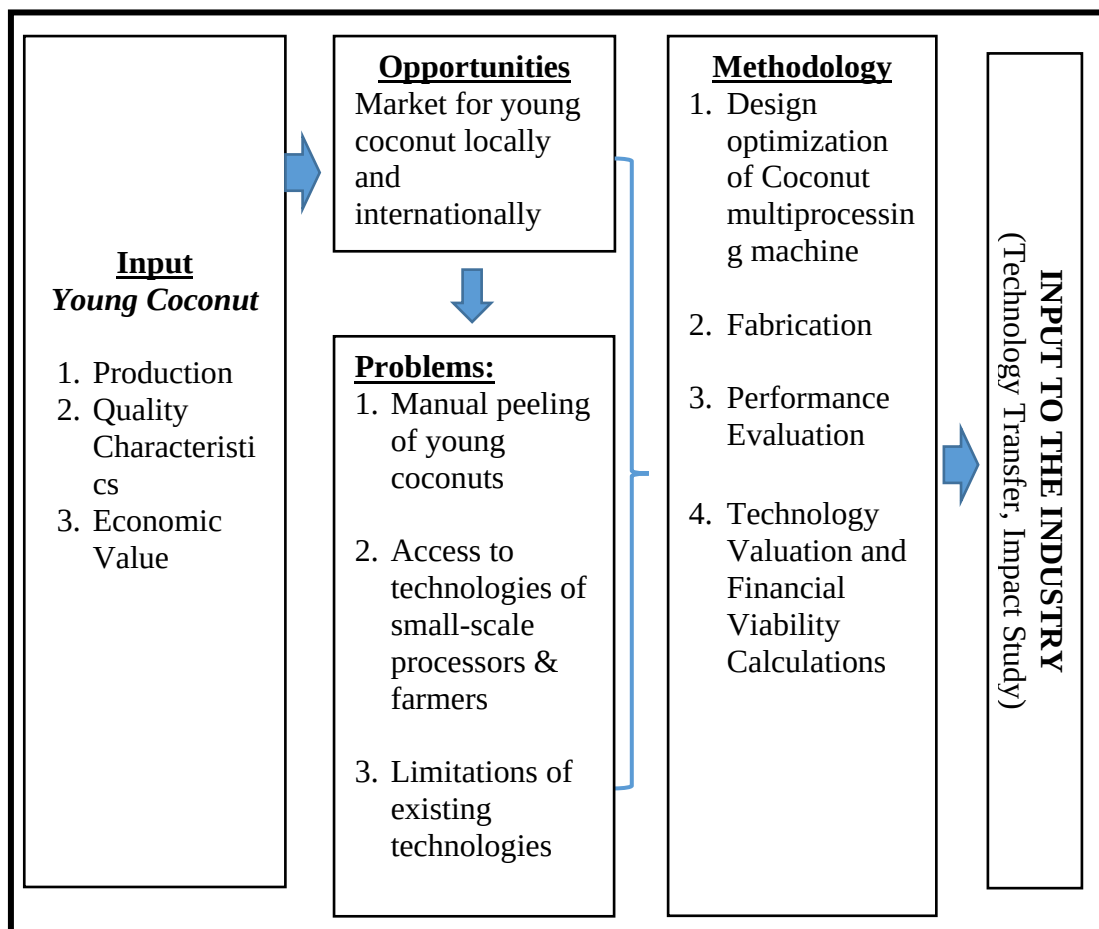
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(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of –the-art of current technology / information from which the project proposal will take-off).

Jarimopas and Pechsamai (2002) have designed a lathe machine to trim coconut. The apparatus pared the rotating fruit by means of a knife that was fed linearly and finely into the fruit by the lathe mechanism. The 70 kg machine featured two sets of knives and was driven by a 0.75 kW single-phase electric motor, which rotated the fruit at 300 rpm. This process could not be considered entirely successful as the fruit required separate manual base cutting. In addition, a green area of about 300 mm in diameter always remained at the crest, which needed to be further trimmed off by a hand.

Nobleza and Borres (2019) designed a young coconut trimming machine to trim coconut into a hexagonal shape. The machine consists of the main frame, side trimming assembly, shoulder trimming assembly, fruit holder/rotating body, and foot pedal assembly. The three basic mechanics employed in this machine's construction are side peeling, top peeling, and bottom cutting, driven by a 0.75-kW single-phase electric motor, which rotated the fruit at 430 rpm. Performance test showed that the trimming machine had a trimming time of 46.81 sec-fruit-1; trimming capacity of 78.41 fruits-hr-1 and trimming efficiency of 93.33%. Furthermore, results showed that after trimming a certain number of coconuts, the blade becomes dull creating a finished product that have uneven texture and an unappealing appearance. Moreover, the machine is only efficient in trimming medium sized-coconuts but not for small and large varieties.

(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested).



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(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

The coconut trimming machine of Nobleza and Borres (2019) will be modified and optimized with the following improvements: (1) provision of a piercing and cutting mechanism to maximize the operations performed by the machine; (2) a peel collection device to promote hygiene, avoid contamination and efficient waste removal (3) optimizing the blade angles to cater varying sizes of coconut; (4) use of blades suited to coconut properties to maintain its sharpness for longer operating period; and (5) elevating the height of the main frame for ease of operation.

Technical Description of the Machine

The coconut multiprocessing machine will comprise of the main frame, shoulder trimming assembly, side peeler assembly, fruit holder, foot pedal assembly, piercing assembly, cutting assembly, and peel collection assembly (Figure 1).

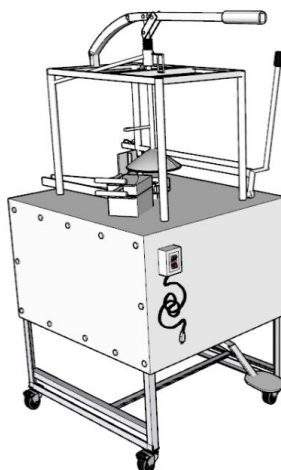


Figure 1. The modified coconut multiprocessing machine.

The **main frame** will support the whole components of the machine. The body will be covered with stainless sheet bounded by bolts. Four identical poles will support the side peeler assembly and shoulder trimming assembly. Metals bars will serve as the frame of the machine. The base cutter positioned at the left side of the frame will be manually controlled by a handle which will be used to cut the lower portion of the coconut fruit mounted in identical pointed edges. The base cutter will be designed in proportion to the height of the fruit holder of the rotating base. A base knife will cut the fruit underside, producing a completely trimmed fruit. The base cutting operation will be mechanically operated but the cutter movement remains manual. For the safety of the operator, the machine is equipped with a switch at front cover which cuts off the electrical supply during loading and unloading of the fruit. The machine will be provided with wheels for portability and easy transport.

The **shoulder trimming assembly** intends to trim the top portion of the coconut. The shoulder blade will be welded to a conic shaped metal to achieve a sharp crest top fruit and will be supported by a tube connecting to the handle. The machine will be provided with compression springs to deflect the handle after the shoulder trimming process. The vertical distance between the shoulder blade and the rotating body holder will be designed according to the maximum height of the coconut fruit with a allowable distance depending on the like while their axis were aligned center to each other. The operator will regulate the up and down movement of the shoulder trimming assembly through the handle.

The **side peeler assembly** will be designed to trim the side portion of the coconut to produce cylindrical shape. It is hanging at the vertically positioned identical poles standing at the main frame. The side peeler blade was

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controlled by a lathe mechanism which provides a linear, smooth and fine cutting motion for the coconut. The handle is responsible for the forward and backward movement of the side peeler assembly.

The rotating base serves as the **fruit holder** of the coconut. The fruit holder will function as a guide and center point that rotates the fruit to its vertical axis during the trimming process. It is driven by an electric motor and will be transmitted by belt and pulleys.

The **foot pedal assembly** will be used as a deflector of the cut coconut base by stepping on the pedal after the trimming process has been done. The shaft enclosed by the rotating tube connects the foot pedal to the husk deflector.

The **piercing assembly** will comprise of a pointed shaft positioned at the top portion of the machine and the up and down action will be controlled by a handle.

A **cutting assembly** will be provided to cut the coconuts into halves. The cutter will be embedded in the base of the frame and will be operated manually.

A **peel collection assembly** will be provided to collect peelings from the shoulder, side and bottom trimmings of the coconut.

Flow of Operation

Positioning of Coconuts. The coconuts will be placed upside down on the rotating base, making sure that the coconut is centered. The conical shoulder trimmer will be used to push and fasten the coconut on the rotating base, securing the stability of the coconut to the fruit holder. A screw lock will be provided to ensure that the shoulder trimmer is firmly fixed.

Side trimming. After setting up the coconut, side trimming will be done by slowly reeling manually the side trimmer to peel the side portion of the coconut to form into a cylindrical shape.

Top Trimming. Top trimming will be done by unlocking the screw of the shoulder trimmer while pulling down the lever, allowing contact of the top blade to the coconut to produce a conic shape.

Base cutting. The bottom cutter will be manually pushed forward into the base of the coconut until the bottom portion will be completely severed from the base. The machine will then be turned off. The foot pedal will then be depressed by stepping on the pedal mechanism to detach the remaining base of the coconut that is still attached on the fruit holder.

Hole piercing. A hole will be punched at the top portion of the peeled coconut to facilitate easy entry of drinking straw.

Cutting. A second cutting blade will be provided to cut the coconut into halves for easy access or removal of coconut meat.

Peel collection. All the peels will be directed to the peel collection chute, situated at the side portion of the machine

Performance Evaluation

The performance of the coconut multiprocessing machine will be evaluated in terms of capacity and efficiency.

Financial Viability

The machine's financial viability will be computed in terms of Investment Cost, Return on Investment (ROI), Break-even Point and Payback Period

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Statistical analysis

The data that will be collected from the study will be analyzed using percentages and means. T-test will be used to compare the performance between the optimized multiprocessing machine and that of the trimming machine by Nobleza and Borres (2019). The level of significance will be set at 5%.

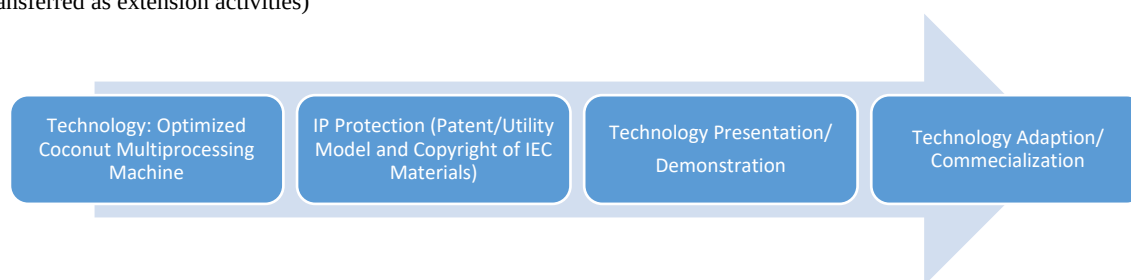
(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart.

Research Objectives	Year 1											
	Jan	Feb	Mar	Apr	May	Jun	July	Aug	Sept	Oct	Nov	Dec
1. Machine Design Modification												
2. Optimization												
3. Performance Evaluation												
4. Technology Valuation												
5. Terminal report writing and finalization												

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

1. Prototype Optimized Coconut Multiprocessing Machine
2. IEC Material /Operations Manual
3. IP protection

(16) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)



(17) Target Beneficiaries (Who the clientele are and what are the expected outcome/effects of the use of the projects output)

Coconut farmers
Restaurants, Hotels and Catering Services

(18) Personnel Requirement

Name	Role (Program, Project, Study Leader)	Field of Specialization
Engr. Marilyn L. Bernabe	Study Leader	Agricultural Machinery
Engr. Arlyn C. Olmo	Co-Researcher	Agricultural Machinery
Dr. Efren L. Linan	Co-Researcher	Environmental Engineering

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Monalyn Oloroso	Co-Researcher	Crop Processing
Ela Villegas	Co-Researcher	Crop Processing
Percent time devoted to the Program/ Project/ Study		
Program/Project Leader (in hours):		
Program/Project Members (in hours):		
(19) Budgetary Requirement		
SOURCES OF EXPENSES	Quantity	Amount (PhP)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)	Research Assistant (1 month)	14,000.00
B. Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)		
Construction and Evaluation Materials		50,000.00
Labor Cost (Construction & Evaluation)		20,000.00
Documentation, photocopy and printing expenses		3,000.00
IP Protection		3,000.00
Communication and Travel		3,000.00
Miscellaneous cost		2,000.00
C. CAPITAL OUTLAY		
Total		95,000.00
(20) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, http://researchjournal.capsu.edu.ph/)		
Jarimopas, B., & Pechsamai, A. (2002). Design and Development of uong coconut peeling machine. Thai Agricultural Research Journal, (20)2, 91-110. Jarimopas, B., and Ruttanadat, N, (2007). Development of a young coconut fruit trimming machine. Department of Agricultural Engineering, Kasetsart University, Kamphaeng Saen Campus, NakohnPathom Thailand. Nobleza, BG and Borres, RL, (2019). Construction and Evaluation of Tender Coconut Husk Trimming Machine. Unpublished Undergraduate Thesis. Capiz State University, Burias Mambusao, Capiz.		
(21) Capsule Curriculum Vitae (One-page brief C.V. only)		